CDL Air Brakes Test Questions and Answers

45 practice questions with answers and explanations, grouped by topic

Written from the Commercial Driver's License Manual (Version 07/20/2026 (form DMV 60A), AAMVA, Virginia DMV edition). Each answer names the manual section it comes from.

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How to use this PDF: answer the questions in Part 1 first, then check yourself with the answer key in Part 2.

Passing score on each CDL knowledge test: at least 80% correct (49 CFR 383.135).

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Part 1: Questions

Air brake system parts (20 questions)

1. What might it mean if the ABS malfunction lamp stays on after the bulb check?

- **A.** The service brakes have failed completely
- **B.** The air tanks need to be drained
- **C.** The spring brakes have been applied
- **D.** You may have lost ABS control at one or more wheels

2. Truck tractors with air brakes built on or after which date are required to have antilock brakes?

- **A.** March 1, 1975
- **B.** March 1, 1994
- **C.** March 1, 1997
- **D.** March 1, 1998

3. Where is the yellow ABS malfunction lamp located on a trailer?

- **A.** On the right side, next to the rear bumper
- **B.** On the left side, at either the front or rear corner
- **C.** On the tractor's instrument panel only
- **D.** On the roof, near the front of the trailer

4. During cold weather, how often should you check the alcohol evaporator container and fill it as needed?

- **A.** Once a week
- **B.** Every day
- **C.** Once a month
- **D.** Only when the low air warning comes on

5. **Your application pressure gauge shows you must use more and more pressure to hold the same speed. What should you do?**
   - **A.** Speed up so airflow cools the brakes
   - **B.** Slow down and use a lower gear
   - **C.** Pump the brake pedal quickly
   - **D.** Pull out the parking brake knob

6. **If the air compressor has its own oil supply, what should you do before driving?**
   - **A.** Drain the compressor oil into the air tanks
   - **B.** Add alcohol to the compressor oil
   - **C.** Check the compressor's oil level
   - **D.** Run the engine at a fast idle for ten minutes

7. **On a vehicle with dual parking control valves, why must you plan carefully when moving it with air from the separate tank?**
   - **A.** It has enough air to release the spring brakes only a few times
   - **B.** It refills only while the engine is turned off
   - **C.** Using it turns off the service brakes
   - **D.** It must also supply air to the trailer brakes

8. **At about what air tank pressure does the governor stop the compressor from pumping air?**
   - **A.** Around 125 psi
   - **B.** Around 55 psi
   - **C.** Around 100 psi
   - **D.** Around 150 psi

9. **On an older vehicle with a front brake limiting valve, what happens when the cab control is set to slippery?**
   - **A.** Air pressure to the front brakes is doubled
   - **B.** Normal air pressure to the front brakes is cut by half
   - **C.** Air pressure to the rear brakes is cut by half
   - **D.** The front brakes are turned off completely

10. **How is a manually operated air tank drain valve opened?**
    - **A.** By pressing the brake pedal several times
    - **B.** By turning it a quarter turn or by pulling a cable
    - **C.** By pulling out the yellow parking brake knob
    - **D.** By running the engine at a fast idle

11. **On newer vehicles with air brakes, what does the parking brake control look like?**
    - **A.** A red, eight-sided push-pull knob
    - **B.** A blue, round push-and-hold button
    - **C.** A yellow, diamond-shaped push-pull knob
    - **D.** A green, square toggle switch

12. What can happen if you press and release the brake pedal when there is no need to?

- **A.** Air can be let out faster than the compressor can replace it
- **B.** The governor raises the cut-out pressure
- **C.** The spring brakes release by themselves
- **D.** The brake linings cool down faster

13. Why should you never push the brake pedal down while the spring brakes are on?

- **A.** It would release the spring brakes and let the vehicle roll
- **B.** It would empty all the air tanks at once
- **C.** It would switch off the antilock brakes
- **D.** The brakes could be damaged by the springs and air pressure acting together

14. At what pressure is the air system safety relief valve usually set to open?

- **A.** 55 psi
- **B.** 100 psi
- **C.** 125 psi
- **D.** 150 psi

15. What is the most common type of foundation brake?

- **A.** The wedge brake
- **B.** The air disc brake
- **C.** The s-cam drum brake
- **D.** The power screw brake

16. Tractor and straight truck spring brakes come fully on when air pressure drops into which range?

- **A.** 20 to 45 psi
- **B.** 55 to 60 psi
- **C.** 80 to 85 psi
- **D.** 100 to 125 psi

17. How does an air brake system turn on the brake lights?

- **A.** With a mechanical switch on the parking brake knob
- **B.** With a sensor on each slack adjuster
- **C.** With an electric switch that works by air pressure
- **D.** With a switch inside the compressor governor

18. How are the supply pressure gauges set up on a vehicle with a dual air brake system?

- **A.** A gauge for each half of the system, or one gauge with two needles
- **B.** One gauge with a single needle for both halves
- **C.** A separate gauge at each wheel's brake chamber
- **D.** No gauge at all, only a warning light

19. If the air compressor stops working, how much braking do the air storage tanks still allow?

- **A.** Enough air for only one light brake application
- **B.** No braking at all, because the tanks empty at once
- **C.** Unlimited braking until the engine is shut off
- **D.** Enough air to use the brakes several times

20. How is a manual reset wig wag different from an automatic wig wag?

- **A.** It rises out of view by itself when pressure goes above 55 psi
- **B.** It sounds a buzzer instead of dropping an arm
- **C.** It drops into view only when pressure goes above 85 psi
- **D.** It has to be put in the out-of-view position by hand

Dual air brake systems (3 questions)

21. Before driving a vehicle with a dual air system, the compressor should build up at least how much pressure in both the primary and secondary systems?

- **A.** 55 psi
- **B.** 85 psi
- **C.** 100 psi
- **D.** 150 psi

22. In a dual air brake system, what happens if one air system is very low on pressure?

- **A.** The other system automatically doubles its pressure
- **B.** Either the front or the rear brakes will not be operating fully
- **C.** The spring brakes release so the vehicle can roll freely
- **D.** Only the trailer brakes stop working

23. The low air warning light and buzzer should shut off when pressure in both systems rises to a value set by the manufacturer. That value must be:

- **A.** Greater than 55 psi
- **B.** Less than 20 psi
- **C.** Between 20 and 45 psi
- **D.** Exactly 45 psi

Inspecting air brakes (9 questions)

24. If brakes with automatic slack adjusters are found to be out of adjustment, what does the manual recommend?

- **A.** Adjust them by hand and keep driving as usual
- **B.** Ignore it, because they will adjust themselves
- **C.** Drain the air tanks and test them again
- **D.** Take the vehicle to a repair facility as soon as possible

25. In a dual air system with the engine at operating rpm, air pressure should build from 85 to 100 psi within:

- **A.** 45 seconds
- **B.** 10 seconds
- **C.** 90 seconds
- **D.** 3 minutes

26. During the engine compartment checks, what should you check on a belt-driven air compressor?

- **A.** The cut-out pressure shown on the dash gauges
- **B.** The thickness of the brake linings
- **C.** The color of the compressor's air filter
- **D.** The condition and tightness of the compressor drive belt

27. **During the walk-around inspection, brake drums or discs must not have cracks longer than:**
    - **A.** One quarter of an inch
    - **B.** One inch
    - **C.** One half the width of the friction area
    - **D.** The full width of the friction area

28. **During the final air brake check, how do you lower tank pressure to see whether the compressor cuts in?**
    - **A.** Run the engine at a fast idle
    - **B.** With the engine idling, step on and off the brake
    - **C.** Rev the engine several times in neutral
    - **D.** Wait with the engine idling and the brakes off

29. **How do you test that the parking brake will hold?**
    - **A.** Stop, put the parking brake on, and pull hard against it in a high gear
    - **B.** Stop, put the parking brake on, and gently pull against it in a low gear
    - **C.** Release the parking brake and let the vehicle roll down a slope
    - **D.** Put the parking brake on and press the brake pedal hard

30. **What is the most common problem found in roadside inspections?**
    - **A.** Out-of-adjustment brakes
    - **B.** Loose air compressor belts
    - **C.** Burned-out ABS malfunction lamps
    - **D.** Leaking alcohol evaporators

31. **When you test the service brakes, about how fast should the vehicle be moving when you apply the brakes firmly?**
    - **A.** About 15 mph
    - **B.** About 25 mph
    - **C.** About 35 mph
    - **D.** About five mph

32. **If the low pressure warning signal does not work and you lose air pressure in a single-circuit air system, what could happen?**
    - **A.** The compressor would pump faster to make up the loss
    - **B.** The service brakes would work normally until you stop
    - **C.** Sudden emergency braking
    - **D.** The brake lights would stay on until you stop

Using air brakes (13 questions)

33. **If you always drive a straight truck or combination with working ABS on all axles, what may you do in an emergency stop?**
    - **A.** Use only the parking brake
    - **B.** Fully apply the brakes
    - **C.** Pump the brake pedal as fast as you can
    - **D.** Shift into neutral before braking

34. At 55 mph on dry pavement, about how much distance does air brake lag add?

- **A.** About 6 feet
- **B.** About 142 feet
- **C.** About 450 feet
- **D.** About 32 feet

35. With the controlled braking method, how hard do you apply the brakes?

- **A.** As hard as you can without locking the wheels
- **B.** All the way, releasing them only when the wheels lock
- **C.** Just hard enough to feel a definite slowdown
- **D.** Lightly at first, then fully after three seconds

36. What happens as overheated brake drums expand?

- **A.** The drums grip the linings harder and brake better
- **B.** The shoes must move farther and press with less force
- **C.** The air pressure in the storage tanks rises
- **D.** The antilock brakes switch off to let them cool

37. Why should you not use the parking brakes when the brakes are very hot?

- **A.** They will freeze so the vehicle cannot move
- **B.** The heat can damage them
- **C.** They will release by themselves after a few minutes
- **D.** They will make the air tanks overfill

38. Why does a heavily loaded vehicle take a long distance to stop when its spring brakes come on?

- **A.** The spring brakes need at least 100 psi to work
- **B.** The ABS turns the spring brakes off
- **C.** The air tanks refill and release the spring brakes
- **D.** The spring brakes do not work on all axles

39. During a normal stop in a vehicle with a manual transmission, when should you push the clutch in?

- **A.** When the engine rpm is down close to idle
- **B.** As soon as you begin pressing the brake pedal
- **C.** Before you begin pressing the brake pedal
- **D.** When the speed drops to about 30 mph

40. In the proper braking technique for a downgrade, about how long should each brake application last?

- **A.** About three seconds
- **B.** About one second
- **C.** About ten seconds
- **D.** About thirty seconds

41. With the proper braking technique on a downgrade, when do you release the brakes?

- **A.** When the vehicle has come to a complete stop
- **B.** When your speed is about 15 mph below your safe speed
- **C.** When your speed is about five mph below your safe speed
- **D.** When the low air pressure warning comes on

42. In stab braking, after you release the brakes, how long can it take for the wheels to start rolling?

- **A.** Up to five seconds
- **B.** Up to ten seconds
- **C.** Up to one second
- **D.** Up to thirty seconds

43. In a vehicle without ABS, what happens when your steering wheels lock up during hard braking on a slippery surface?

- **A.** The vehicle stops in a shorter distance
- **B.** The ABS malfunction lamp comes on
- **C.** You lose steering control
- **D.** The spring brakes come on automatically

44. When only the tractor has ABS, what should you do if the trailer begins to swing out?

- **A.** Apply the brakes fully and hold them
- **B.** Pull out the parking brake knob
- **C.** Speed up to pull the trailer straight
- **D.** Let up on the brakes, if you can safely do so

45. If your brakes are wet, how should you dry them?

- **A.** Set the parking brakes at every stop
- **B.** Drive in a high gear without using the brakes
- **C.** Use the brakes lightly while driving in a low gear
- **D.** Drain the air tanks more often

Part 2: Answer key

Each answer has a short explanation in our own words and the CDL manual section it comes from. Reread that section for every question you missed.

Air brake system parts

1. **D. You may have lost ABS control at one or more wheels**
   A lamp that does not go out after start-up, or that lights while driving, warns that ABS may not be working at one or more wheels.
   Manual section 5.1.16 Antilock Braking Systems (ABS)

2. **C. March 1, 1997**
   The ABS requirement began for air-braked truck tractors built from March 1, 1997, one year earlier than for other air brake vehicles such as trucks, buses and trailers.
   Manual section 5.1.16 Antilock Braking Systems (ABS)

3. **B. On the left side, at either the front or rear corner**
   Tractors, trucks and buses show the ABS lamp on the dash, while trailers carry it on the left side at a front or rear corner.
   Manual section 5.1.16 Antilock Braking Systems (ABS)

4. **B. Every day**
   In cold weather the alcohol supply should be checked and topped up daily so it keeps working against ice in the air system.
   Manual section 5.1.5 Alcohol Evaporator

5. **B. Slow down and use a lower gear**
   Needing more application pressure for the same speed is a sign of brake fade, so you should reduce speed and shift to a lower gear.
   Manual section 5.1.10 Application Pressure Gauge

6. **C. Check the compressor's oil level**
   A compressor with a separate oil supply is not lubricated by engine oil, so its own oil level has to be checked before each trip.
   Manual section 5.1.1 Air Compressor

7. **A. It has enough air to release the spring brakes only a few times**
   The extra tank lets you release the spring brakes in an emergency, but its air runs out after a few uses, which could leave you stopped in a dangerous spot.
   Manual section 5.1.15 Parking Brake Controls

8. **A. Around 125 psi**
   The cut-out level is about 125 psi; at that point the governor stops the compressor, and it lets pumping start again only when pressure falls to the cut-in level.
   Manual section 5.1.2 Air Compressor Governor

9. **B. Normal air pressure to the front brakes is cut by half**
   The slippery setting halves the air going to the front brakes, which lowers the vehicle's stopping power, so the control should be kept at normal.
   Manual section 5.1.13 Front Brake Limiting Valve

10. **B. By turning it a quarter turn or by pulling a cable**
    Manual drain valves at the bottom of each tank are worked by hand, either with a quarter turn of the valve or with a pull cable.
    Manual section 5.1.4 Air Tank Drains

11. **C. A yellow, diamond-shaped push-pull knob**
    Newer air brake vehicles use a yellow diamond knob that is pulled out to set the spring brakes and pushed in to release them.
    Manual section 5.1.15 Parking Brake Controls

12. **A. Air can be let out faster than the compressor can replace it**
    Every release of the pedal lets some compressed air escape, so needless pumping can drain the tanks faster than the compressor refills them, and the brakes stop working if pressure gets too low.
    Manual section 5.1.7 The Brake Pedal

13. **D. The brakes could be damaged by the springs and air pressure acting together**
    With the springs already applying the brakes, adding air pressure from the pedal can overload and damage them, and not every system is built to prevent this.
    Manual section 5.1.15 Parking Brake Controls

14. **D. 150 psi**
    The safety valve guards the first tank and the rest of the system against too much pressure, and it normally opens at 150 psi.
    Manual section 5.1.6 Safety Valve

15. **C. The s-cam drum brake**
    Foundation brakes sit at each wheel, and the s-cam drum brake is the type used most; wedge and disc brakes are less common.
    Manual section 5.1.8 Foundation Brakes

16. **A. 20 to 45 psi**
    Spring brakes apply fully somewhere between 20 and 45 psi, most often between 20 and 30 psi, but you should stop safely as soon as the low air warning comes on rather than wait for this.
    Manual section 5.1.14 Spring Brakes

17. **C. With an electric switch that works by air pressure**
    Air pressure from a brake application operates an electric switch, which lights the brake lights to warn drivers behind you.
    Manual section 5.1.12 Stop Light Switch

18. **A. A gauge for each half of the system, or one gauge with two needles**
    With a dual system the driver needs to see the pressure in each half, so there are two gauges or a single gauge showing two needles.
    Manual section 5.1.9 Supply Pressure Gauges

19. **D. Enough air to use the brakes several times**
    The storage tanks keep a reserve of compressed air, so the brakes can still be applied a number of times after a compressor failure.
    Manual section 5.1.3 Air Storage Tanks

20. **D. It has to be put in the out-of-view position by hand**
    An automatic wig wag lifts out of view on its own once pressure is above 55 psi, but the manual reset type must be moved back by the driver, and it will not stay up until pressure is above 55 psi.
    Manual section 5.1.11 Low Air Pressure Warning

Dual air brake systems

21. **C. 100 psi**
    You should wait until both halves of a dual system have at least 100 psi before you start driving.
    Manual section 5.2 Dual Air Brake

22. **B. Either the front or the rear brakes will not be operating fully**
Each half of a dual system serves different axles, so losing pressure in one half weakens the front or rear brakes and makes stopping take longer.
Manual section 5.2 Dual Air Brake

23. **A. Greater than 55 psi**
The manufacturer can choose the shut-off point for the warning, but it has to be above 55 psi.
Manual section 5.2 Dual Air Brake

Inspecting air brakes

24. **D. Take the vehicle to a repair facility as soon as possible**
An automatic adjuster that is out of limits usually points to a mechanical problem, and adjusting it by hand tends to hide that problem rather than fix it, so the vehicle should be repaired soon.
Manual section 5.3.2 During Step 5 Walk-around Inspection

25. **A. 45 seconds**
A dual air system should go from 85 to 100 psi in no more than 45 seconds at operating rpm, unless larger-than-minimum tanks allow a longer time under the manufacturer's specifications.
Manual section 5.3.3 Step 7 Final Air Brake Check

26. **D. The condition and tightness of the compressor drive belt**
When the compressor is driven by a belt, the engine compartment check includes making sure that belt is in good shape and properly tight.
Manual section 5.3.1 During Step 2 Engine Compartment Checks

27. **C. One half the width of the friction area**
A crack in a drum or disc is acceptable only if it is shorter than half the width of the friction area.
Manual section 5.3.2 During Step 5 Walk-around Inspection

28. **B. With the engine idling, step on and off the brake**
Pressing and releasing the brake pedal uses up air, so the gauge falls until the governor should let the compressor start pumping again.
Manual section 5.3.3 Step 7 Final Air Brake Check

29. **B. Stop, put the parking brake on, and gently pull against it in a low gear**
The test is a gentle pull in a low gear with the parking brake set, which shows whether the parking brake can keep the vehicle from moving.
Manual section 5.3.3 Step 7 Final Air Brake Check

30. **A. Out-of-adjustment brakes**
Brakes that are out of adjustment show up more than any other problem at roadside inspections, which is why checking the slack adjusters matters.
Manual section 5.3.2 During Step 5 Walk-around Inspection

31. **D. About five mph**
The service brake test is done at a slow roll of about five mph, while you watch for pulling to one side, an odd feel or a slow stop.
Manual section 5.3.3 Step 7 Final Air Brake Check

32. **C. Sudden emergency braking**
Without a working warning you would not notice falling air pressure, and in a single-circuit system that can lead to the brakes coming on suddenly.
Manual section 5.3.3 Step 7 Final Air Brake Check

Using air brakes

33. **B. Fully apply the brakes**

The one exception to braking as you always have is a vehicle with working ABS on every axle, where full brake application is allowed in an emergency stop.

Manual section 5.4.2 Braking with Antilock Brakes

34. **D. About 32 feet**

Because air takes time to reach the brakes, brake lag adds roughly 32 feet at 55 mph, on top of perception, reaction and braking distance.

Manual section 5.4.4 Stopping Distance

35. **A. As hard as you can without locking the wheels**

Controlled braking means braking at the edge of wheel lockup while keeping steering movements very small.

Manual section 5.4.3 Emergency Stops

36. **B. The shoes must move farther and press with less force**

Heat makes the drums grow larger, so the shoes travel farther and press with less force, which is one way brakes fade.

Manual section 5.4.5 Brake Fading or Failure

37. **B. The heat can damage them**

Setting the parking brakes on very hot brakes, such as right after a steep grade, can cause heat damage, so chock the wheels on level ground and let the brakes cool first.

Manual section 5.4.8 Parking Brakes

38. **D. The spring brakes do not work on all axles**

Spring brakes act on only some of the axles, so a heavy load needs a long distance to stop, and it is safer to stop while there is still air for the foot brake.

Manual section 5.4.7 Low Air Pressure

39. **A. When the engine rpm is down close to idle**

Keep the clutch engaged while braking until engine speed has fallen almost to idle, then select a starting gear once stopped.

Manual section 5.4.1 Normal Stops

40. **A. About three seconds**

Each firm application to bring speed down by about five mph should take roughly three seconds.

Manual section 5.4.6 Proper Braking Technique

41. **C. When your speed is about five mph below your safe speed**

You brake until speed falls roughly five mph under your safe speed, let off, and brake again once speed climbs back up to the safe speed.

Manual section 5.4.6 Proper Braking Technique

42. **C. Up to one second**

The wheels may need as long as a second to roll again, and reapplying the brakes before they do will keep the vehicle from straightening out.

Manual section 5.4.3 Emergency Stops

43. **C. You lose steering control**

Locked front wheels cannot steer the vehicle, so locking them up takes away your ability to steer.

Manual section 5.4.2 Braking with Antilock Brakes

44. **D. Let up on the brakes, if you can safely do so**

Tractor-only ABS helps you keep steering and lowers the chance of a jackknife, but you still need to watch the trailer and ease off the brakes when it starts to swing.

Manual section 5.4.2 Braking with Antilock Brakes

45. **C. Use the brakes lightly while driving in a low gear**

Light braking in a low gear heats the brakes and dries them out.

Manual section 5.4.8 Parking Brakes

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